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・講演タイトル Pedestrian bidirectional flow: a case study

・アブスト内容(約 500 字)

This study presents an empirical observation of a bidirectional flow inside a crowded subway station in central Tokyo. Based on the flow measured in a corridor-like section in which pedestrians walk in opposite directions, a flow-regime diagram has been built which allows predicting the type of flow observed based on the amount of pedestrians walking in each direction. For each type of flow (free-flow, congested motion and deadlock) typical properties were found allowing an accurate distinction between each phenomena. A critical analysis on the data obtained revealed that most of the deadlocks were associated with simultaneous train arrivals on different platforms. A fairly limited delay in arrival time (15 s) is considered being sufficient for avoiding deadlock formation.

Later, the experimental data were used to calibrate and validate a simulation model specifically designed to deal with heavily congested scenarios. A good agreement was found between experimental and numerical data, thus allowing an in-depth analysis of pedestrian behavior. This revealed that body rotation and changes in direction can help dissolving the deadlocks once formed.

Results from this study may help designing infrastructures accommodating large crowds and prevent the occurrence of accidents in the future.